

The Impact of Climate Change on Harmful Algal Blooms
(HABs) in the Genesee-Finger Lakes Region: A Review
of Trends and Projections Across Nine New York
Counties

Paul A. Shannon

Climate Solutions Accelerator

Rochester, NY

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1. Abstract

The Genesee-Finger Lakes Region (G/FLR) is an expansive region in Western New York State consisting of nine counties. This region, home to the 11 Finger Lakes and one Great Lake, is regularly impacted by runoff-driven eutrophication and subsequent HAB events (NYSDEC, 2001). It has been established in the literature that, as an overall trend, increasing levels of climate change exacerbate HAB incidence, severity, and levels of algal growth (CDC, 2024). However, to the best of the author's knowledge, no regional study to date has explored the problem of climate change-fueled HAB incidence in the G/FLR specifically. Between 2020 and 2024, HAB reports in the region increased sixfold, with individual report submissions rising from an average of 36 to over 220 annually. Three counties—Yates, Seneca, and Ontario—emerged as clear Algal Hotspots, each demonstrating report rates more than 40 times higher than neighboring counties. These same counties showed seasonal temperature averages consistently within the *Microcystis*-favorable range (20–30°C) and maintained high precipitation anomalies, which strongly correlated ($r = 0.77$) with HAB frequency. Soil analysis further revealed that lakes classified as “High-Severity Problem Lakes” (Canandaigua, Seneca, and Cayuga) are surrounded primarily by Group D and B/D soils—those most prone to runoff.

2. Introduction

Harmful algal blooms, otherwise known as HABs, are widespread environmental phenomena characterized by an overabundance of certain types of algae-like organisms, primarily cyanobacteria, but including some kinds of dinoflagellates and/or diatoms as well (Cornell Wildlife Health Lab, 2023). HABs have increasingly affected fresh and saltwater ecosystems globally and, possibly in part due to increased monitoring efforts, across the United States as well (Hallegraeff et al., 2021).

There have been notable increases in HAB incidence within the G/FLR of New York over the past decade (Johnston et al., 2024). For example, in 2017, all 11 Finger Lakes experienced open water, shoreline, or both types of HABs (Johnston et al., 2024). The Finger Lakes are especially susceptible to certain HAB events due to their historical glacial formation, contributing to their depth(s), ranging from 30-650 ft. (Wiltse, 2022; NASA, 2013). Deeper bodies of water such as the Finger Lakes are affected by processes like stratification, ultimately contributing to favorable HAB conditions by trapping nutrients within the warmer surface layer(s), increasing algal food availability (SGWASA, 2024). Shallower, coastal areas of these bodies of water are additionally susceptible, as there is a more direct impact by sunlight, leading to potentially higher water temperatures within the water column.

Although a multitude of factors contribute to the formation of HABs, they

are known to form in nearly any kind of warm, slow—or non-moving—body of water with an excess of nutrients such as NO_3 and PO_4 (Department of Environmental Protection, n.d.). In terms of temperature, most cyanobacterial strains maintain an “ideal” temperature range of between 27°C and 37°C , though anything over 25°C is considered favorable (Hu et al., 2023). The overabundance of nutrients observed within HAB events is called eutrophication, and is generally classified as any nutrient loading which exceeds the body of water’s self-purifying capacity (NOAA, 2023).

HABs are known to form a surface scum atop the surface of afflicted bodies of water, often described as similar to spilled paint or pea soup in appearance. They can manifest in a variety of colors depending on the HAB-causing strain, but are most commonly blue, green, red, or brown (CDC, 2024). Algal scum is extremely dense, and can therefore stop the sun’s rays from penetrating the surface of the water. This may halt photosynthetic activity within the water and can ultimately induce hypoxia (NCCOS, 2024), leading to adverse environmental effects.

Hypoxia is characterized by low levels of dissolved O_2 within the water column ($\text{O}_2 < 3 \text{ mg/L}$), and can contribute to ecological events such as fish kills (Small et al., 2014). Hypoxic events are additionally detrimental to recreational, commercial, and other economic industries when they occur, on both local and global scales (e.g. fishing, swimming, boating, dining).

Eutrophication most commonly arises due to a multitude of factors, but

arguably most significantly agricultural runoff. In terms of applied fertilizers used on farmland, surface runoff has been responsible for significant nutrient loading in recent years. In 2022 alone, according to the USDA, the United States applied 11.96 million metric tons of nitrogen fertilizer to farmland (1.19×10^{10} kilograms). The literature has shown that, on average, 20% of applied nitrogen-based fertilizers are washed away following application due to this phenomenon (WRI, n.d.). Therefore, it is reasonable to hypothesize that extensive amounts of nitrates, nitrites, and other fertilizer compounds are being introduced into the water column on an annual basis nation-wide, only exacerbating the current problem seen with runoff.

Eutrophication contributes to the very “ideal” water conditions in which cyanobacteria thrive. The phenomenon takes susceptible bodies of water (i.e. slow/non-moving, warm water sources) and exacerbates their already existing predisposition to algal formation, by providing an excess of an algal “food source” in the form(s) of phosphorus and/or nitrogen.

HABs—particularly those which consist of cyanobacteria—are known to produce numerous bioactive metabolites, referred to as cyanotoxins. There are three primary categories of these toxins, largely formatted based on their mode(s) of action: neurotoxins, hepatotoxins, and dermatotoxins. Some of the most frequently-seen cyanotoxins produced by HABs are a classification of hepatotoxins known as the Microcystins, produced by *Microcystis* algal strains, of which over

300 congeners exist. Microcystis-LR (MC-LR), is a potent liver tumor promoter, encourages apoptosis through cytotoxic means, and is among one of the most common MCs encountered in the natural environment (Ito, 1997; Lankoff et al., 2004; Chaffin et al., 2023).

Climate change can exacerbate eutrophication, as it creates conditions which can increase nutrient loading in water, such as elevated temperatures, which can ultimately release deposited nutrients from lake sediments (Rodgers, 2021). It has been demonstrated that climate change may impact HAB incidence; however, to the best of the author's knowledge, the topic of research has not yet been described in-depth with focus on the G/FLR, with many studies instead focusing on universal algal mechanisms themselves. As such, the intended purpose of this paper is to i.) assess both current regional climatic trends and future predictions relating to "HAB-ideal" conditions; ii.) determine to which extent nutrient leaching may occur in the area, which exacerbates HAB formation; and iii.) determine which region(s), if any, pose the most significant algal threats.

3. Methodology

3.1: "Problem Lakes" Identification and Definition

"Problem Lakes" were identified from collected DEC HAB data, and formatted into one of the following categories: 1.) Low Priority; 2.) Low-Severity; 3.) Mid-Severity; and 4.) High-Severity. These classifications were assigned based on i.) how often HAB events are observed and ii.) the number of annual reports

submitted to DEC per water body. A Low-Severity Problem Lake averages 2.0-9.9 reports, a Mid-Severity averages 10.0-49.9 reports, and High-Severity blooms average 50+ reports. If one particular lake maintained less reports than a High-Severity bloom, for instance, but experiences significant cyanotoxin release, it would be classified as High-Severity. Any lake assigned a “Low Priority” classification is not considered a long-term Problem Lake, even if it has experienced HAB events in the past, due to a lack of recurrence and/or toxin prevalence.

3.2: Soil Hydrological Group Determination

The soil hydrologic groups for each body of water determined to be “Low-Severity” or worse was determined using soil data sourced from the Web Soil Survey (WSS), an extensive database provided by the National Cooperative Soil Survey and operated by the USDA Natural Resources Conservation Service. For each lake assessed, detailed soil maps—such as the example provided in *Figure 1*—were analyzed. These maps illustrate varying soil units within the watershed boundaries, each assigned a specific hydrological classification (A, B, C, D, or dual classifications such as B/D or C/D). Soil groups were interpreted based on their infiltration rates and runoff potential, with Group A representing

soils with high infiltration and low runoff potential, and Group D indicating soils with slow infiltration and high runoff risk. By examining the acreage and proportional coverage of each soil type within each lake's watershed, the dominant hydrological group—and thus the runoff potential—was identified.



Figure 1 depicts a representative

section from the Conesus Lake watershed analysis,

Figure 1: WSS plot analysis; each "puzzle piece" represents a plot of a different hydrologic group and/or soil type. Image provided is from Conesus Lake analysis.

highlighting soil makeup of the surrounding area (shown through various colors) encountered during this assessment.

4. Results

Between 2020 and 2024, the nine counties within the Genesee-Finger Lakes Region (G/FLR) have experienced a notable rise in both HAB occurrences and individual HAB reports. Over these years, the mean number of water bodies reported to the DEC yearly has increased from approximately 1.67 to 2.67. Individual annual reports also surged dramatically, over six-fold, from an average of 36 to 222.

Notably, Yates, Seneca, and Ontario counties consistently reported significantly higher occurrences than the remaining six counties, averaging 243 annual reports over four years, compared to just 6 yearly in the others ($p < 0.05$). These counties additionally reported approximately 1.7-fold (± 0.30) more affected water bodies. Due to these marked increases, Yates, Seneca, and Ontario counties have been identified as potential "Algal Hotspots" (see *Table 1* & *Figure 2* for detailed data).

Year	Mean Algal "Non-Hotspot" Reports	Mean Algal "Hotspot" Reports
2020	6.333333333	96
2021	5	191
2022	6	125.3333333
2023	6.5	153
2024	9.166666667	648.3333333
Year	Algal "Non-Hotspot" Water Bodies	Algal "Hotspot" Water Bodies
2020	1.6	2.333333333
2021	1.5	3
2022	2	3.666666667
2023	2.166666667	3
2024	2	4

Table 1: "Algal Hotspots" vs. Unproblematic Counties

Precipitation Anomaly, a representation of how much excess rainfall is occurring compared to the pre-established 100-year NOAA average, showed a possibly strong Pearson Product-Moment correlation coefficient with the number of individual HAB reports within the region, at a value of 0.77. The number of individual reports and the number of reported bodies of water maintained smaller

correlation coefficients with the population value (<-0.40 and <-0.05 , respectively).

For each HAB-afflicted lake (as sourced from DEC data), the severity of HAB events were estimated. Overall, four (4) lakes were classified as "Low-Severity Problem Lakes" (Conesus Lake, Hemlock Lake, Java Lake, and the Monroe County portion of Lake Ontario). Two (2) bodies of water, Honeoye Lake and Keuka Lake, were determined to be possible "Mid-Severity Problem Lakes". Three (3) lakes, Cayuga, Seneca, and Canandaigua, were classified as "High-Severity" Problem Lakes, with considerably higher 5-Year Report Averages and Peaks than all others. The final three (3) referenced High-Severity water bodies are much grander in impact than others considered, with 5-Year Report Averages of 107, 189, and 113, respectively. Any body of water classified as "Low Priority" was considered, and ultimately, despite potential periodic HAB events, were hypothesized to not present a long-term severe problem.

Algal "Hotspot" vs. "Non-Hotspot" Individual Reports

Counties deemed "problematic" vs. those deemed "non-problematic"

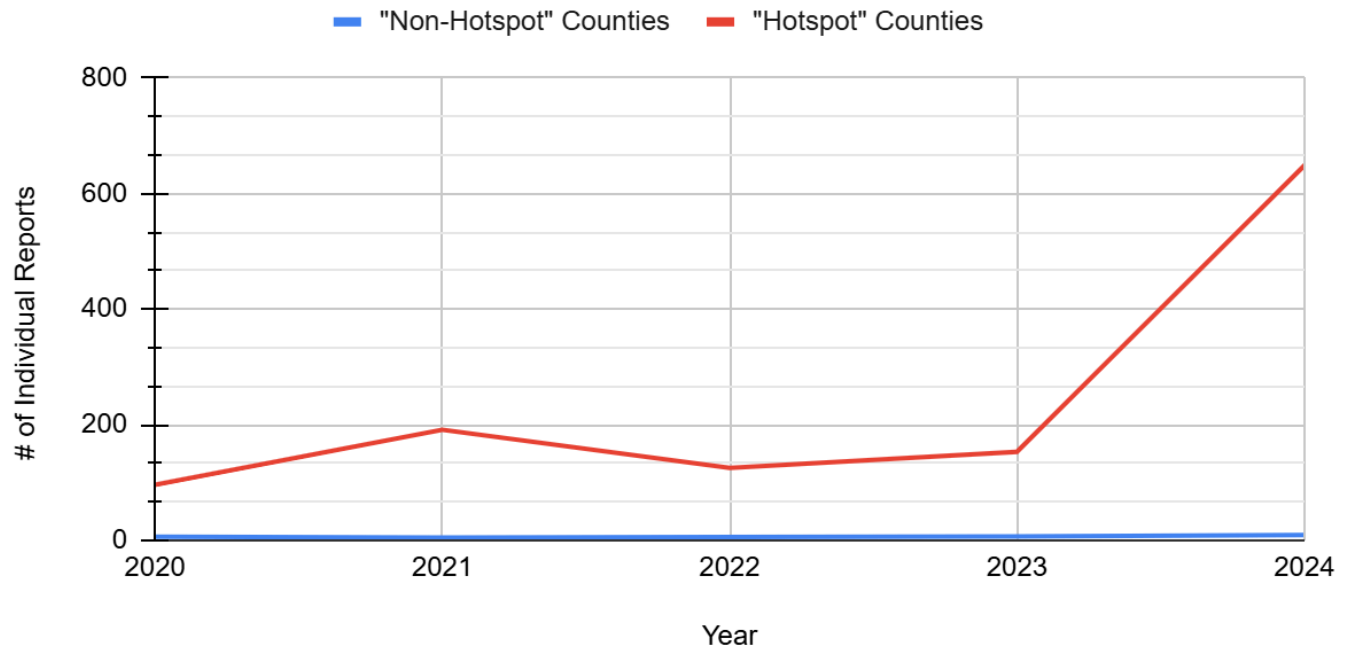


Figure 2: "Algal Hotspots" vs. Unproblematic Counties in terms of individual HAB reports submitted to DEC.

Algal "Hotspot" vs. "Non-Hotspot" Bodies of Water

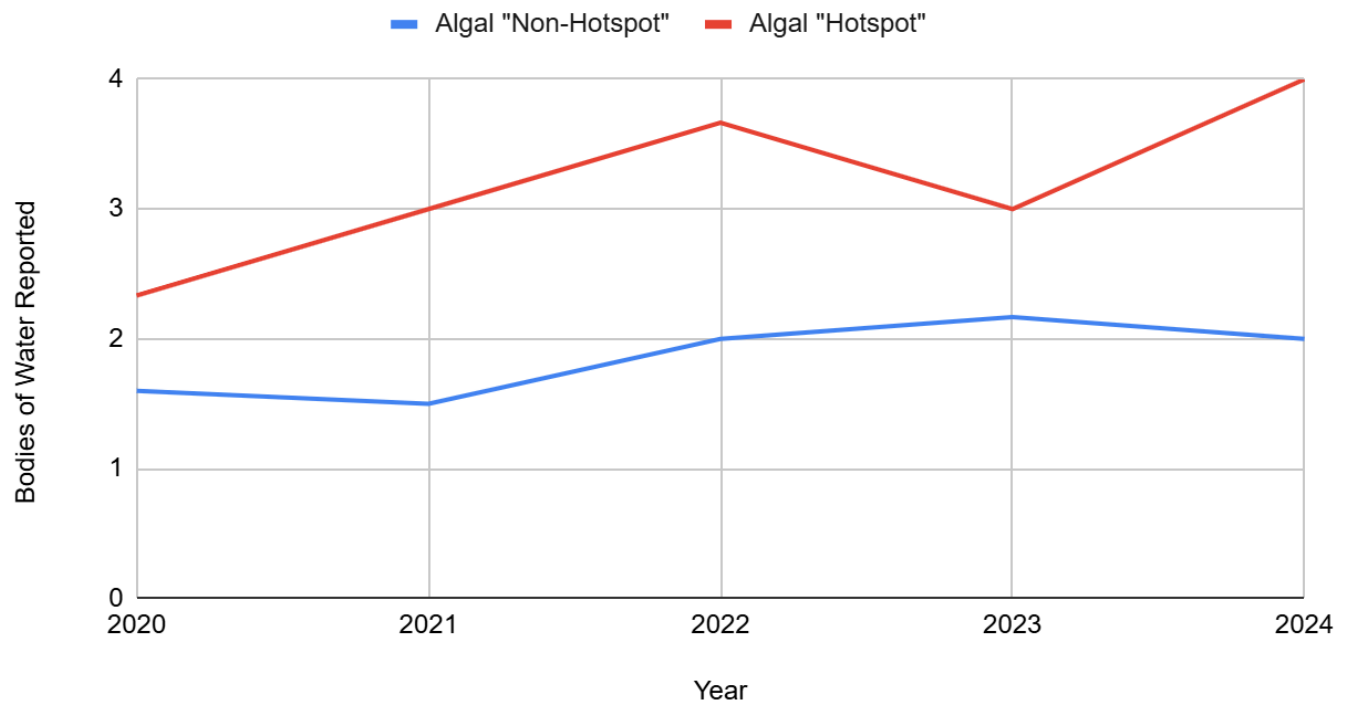


Figure 3: "Algal Hotspots" vs. Unproblematic Counties in terms of # of reported bodies of water submitted to DEC.

Counties	Water Body Name	5-Year Report Average	5-Year Report Peak	Lake "Problem" Classification	Major Soil Hydrologic Group(s)	Runoff Potential as per Soil Group
Livingston	Conesus Lake	7.4	13	Low-Severity Problem Lake	B/D & D	Moderate-High
Livingston	Hemlock Lake	3	5	Low-Severity Problem Lake	B/D & D	Moderate-High
Livingston	Avon Marsh Dam Pond	1.8	5	Low Priority	N/A	N/A
Monroe	Lake Ontario (Monroe)	8.2	24	Low-Severity Problem Lake	B/D, C/D, D	Moderate-High
Monroe	Irondequoit Bay	0.6	1	Low Priority	N/A	N/A
Monroe	Braddock Bay	0.6	2	Low Priority	N/A	N/A
Monroe	Long Pond	0.4	2	Low Priority	N/A	N/A
Ontario	Honeoye Lake	42.2	55	Mid-Severity Problem Lake	Heavily D & B/D, some A/B	High (if undrained)
Ontario, Yates	Canandaigua Lake	113.2	292	High-Severity Problem Lake	Mix of B, B/D, and D soils	High (if undrained)
Ontario	Castle Creek	0.4	2	Low Priority	N/A	N/A
Ontario	Fairport Reservoir	0.2	1	Low Priority	N/A	N/A
Seneca	Cayuga Lake	107	147	High-Severity Problem Lake	Prevalent B/D, C/D, D soils, some A/B	High (if undrained)
Seneca, Yates	Seneca Lake	189	377	High-Severity Problem Lake	Prevalent B/D, C/D, D soils, some A/B	High (if undrained)
Wayne	Port Bay	0.6	2	Low Priority	N/A	N/A
Wayne	Sodus Bay	1	3	Low Priority	N/A	N/A
Wyoming	Java Lake	4.2	7	Low-Severity Problem Lake	Mix of A, B/D, and D soils	Moderate-High (if undrained)
Wyoming	Silver Lake	1	5	Low Priority	N/A	N/A
Yates	Keuka Lake	13.4	16	Mid-Severity Problem Lake	Extensive D & B/D	High

Table 2: All 18 major, named, recurring bodies of water reported to DEC's HAB Report System 2020-2024. Color-coded based on "Problem Severity" (i.e. likelihood of returning, number of reports submitted). Runoff potential determined based on hydrologic groups.

In recent years, (2020-2024), average seasonal temperatures in “Algal Hotspots” Yates, Seneca, and Ontario counties remained consistently within the favorable growth range for *Microcystis* strains (20–30°C), with each year supporting conditions which may potentially exacerbate bloom formation. In all three counties, each year examined (2020-2024) fell within the strain's ideal range, with no recorded average exceptions since 2009. Conversely, the optimal temperature range for Cylindrospermopsin-producing algae (25–30°C) was not met in Ontario County at all during the same period, which is notable given the past presence of *Dolichospermum* algae. Although average temperatures trended upward, values remained below this threshold, with a maximum of 23.4°C in 2020.

5. Discussion

The G/FLR encompasses an area of approximately 4,575 mi² of land and is home to over 1.2 million people within 188 municipalities as per the 2010 U.S. census, as well as the third largest city in all of New York, Rochester (G/FLRPC, n.d.). It is a distinct region in Western NY, made up of 9 counties (Genesee, Livingston, Monroe, Orleans, Ontario, Seneca, Wayne, Wyoming and Yates).

This region, and these nine particular counties were chosen due to an array of factors: the area contains deep, narrow “finger” lakes (such as Seneca and Cayuga) as well as shallower bodies (e.g. parts of Skaneateles and Owasco), and these locations respond directly to nutrient loading. There is additionally a good representative sample, in terms of environmental diversity, in the region. The

G/FLR houses both urbanized areas with more complex nutrient sources, like Genesee, as well as predominantly rural counties where agricultural influences would be more pronounced, such as Seneca.

According to data from NYSDEC's *Citizens Statewide Lake Assessment Program* (CSLAP), specifically their *2018 Water Quality Report*, it shows that, within the same year that 10/11 of the Finger Lakes exhibited blooms, a few distinctive trends could be observed, with one outstanding: phosphorus levels maintained a strong positive correlation with chlorophyll-a, reinforcing the role of phosphorous presence in bloom formation. This encourages some degree of action to diminish regional nutrient loading.

There was an observed significant increase in HAB reports between 2020 and 2024, particularly within Yates, Seneca, and Ontario counties, which have emerged as distinct "Algal Hotspots." The significantly higher average number of reports in these three (3) counties compared to the other six (6) counties within the region indicates one, or a combination of, the following factors: 1.) geographical or anthropogenic discrepancies; 2.) increased HAB severity in "Problem Lakes"; 3.) varying weather patterns across regions.

Precipitation anomalies, representing deviations in rainfall from established historical averages, exhibited a notably strong correlation coefficient (~0.77) with HAB report frequency, contributing to the scientific consensus that increased rainfall—a predicted effect of climate change—is in fact likely exacerbating

nutrient runoff, thus fueling eutrophication and subsequent HAB events. Counties with higher rainfall anomalies, specifically Yates and Seneca, correspondingly reported higher frequencies of HABs, strengthening the argument that intensified precipitation events directly amplify nutrient runoff and exacerbate algal growth conditions. This may also emphasize the hydrological soil results, given their association with runoff.

According to EPA and AAPFCO data, New York State purchased an average of $78,000,000 \pm 16,650,000$ kg of N-based fertilizer annually from the years 2011-2017. The reported percent change (44%) for these years is significant compared to years prior (2002-2006), shown to be -11%. In the same timespan, NYS purchased an average of $28,266,000 \pm 4,703,000$ kg of P-based fertilizer, with the percent change from 2002-2011, reported as -20%, rising to 28% from 2011-2017. This indicates an ongoing increasing trend in the magnitude of fertilizer purchased as time has increased, though more exponential in terms of N-fertilizer than P-based fertilizers. It may therefore be inferred that current values are even larger. As—on average—20% of N-based fertilizers and up to 2% of P-based fertilizers are lost to agricultural runoff, if all purchased product was, at some point applied, New York State would have released over 100,000,000kg of purchased nitrogen fertilizer and 2,000,000 - 4,000,000kg of purchased phosphorus fertilizer in these six years alone.

Analysis of soil hydrologic groups further clarified regional susceptibility.

High-Severity Problem Lakes (Canandaigua, Seneca, and Cayuga Lakes) predominantly featured Group D soils, indicative of poor drainage and high runoff potential. This soil profile amplifies eutrophication risks due to enhanced nutrient loading following precipitation events. Additionally, Mid-Severity Problem Lakes (Honeoye and Kekua Lakes) showed similar soil runoff potential, reinforcing the relationship between regional soil characteristics and HAB vulnerability.

6. Conclusion

This report, ultimately, demonstrates the association between actively changing climatic conditions, ongoing nutrient dynamics, and increased HAB occurrence within the G/FLR. The findings suggest that rising precipitation values—amplified by regional climate change—are likely contributing to elevated levels of nutrient runoff into surface waters, particularly in areas with hydrologically "unfavorable" soils. "High-Severity Problem Lakes" such as Seneca, Cayuga, and Canandaigua are consistently surrounded by Group D and B/D soils, which possess limited infiltration capacity and high runoff potential. When coupled with documented increases in nitrogen and phosphorus fertilizer application across the state, the likelihood of excess nutrient delivery to aquatic systems is significantly enhanced. This creates the ideal eutrophic conditions for increased HAB incidence.

Yates, Ontario, and Seneca counties emerged as distinct "Algal Hotspots," displaying disproportionately high HAB report frequencies between 2020 and

2024. These trends were not mirrored in the remaining counties, suggesting a geographically unequal distribution of climate-induced bloom severity and a potentially uneven future impact. It is also important to note that seasonal temperature averages across the region, particularly within hotspot counties, has remained well within the optimal range for *Microcystis* growth for sixteen years straight. This temperature consistency—paired with evidence of stratification in deeper lakes—further supports the argument that the G/FLR is increasingly predisposed to HAB formation under ongoing climate trajectories.

To ensure and to maintain ecological longevity, public health safety, and to minimize disruptions to local industries, the implementation of informed mitigation strategies is critical, as well as ensuring education regarding HABs on a local stage. This includes enhanced year-round monitoring of bloom toxins and runoff mitigation efforts such as riparian buffer zones, wetland reintroduction, and agricultural BMPs. Further research should continue to integrate HAB monitoring with climate projections and watershed-scale nutrient data, in order to more accurately anticipate emerging risk zones. Moreover, while it is true that the solution to HABs likely lies in education, further research should be done regarding remediation of these risk zones on a regional scale. Otherwise, HAB frequency and intensity in the G/FLR will likely be exacerbated, placing increasing pressure on aquatic ecosystems, drinking water resources, and regional economies all the same.

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